



IMAGE PROCESSING & COMPUTER VISION

RTI235007

Minggu 6: Filter Spasial

TEACHING TEAM
MATA KULIAH PENGOLAHAN CITRA DAN VISI KOMPUTER

Outline

- ❖ Filter
- ❖ Convolution
- ❖ Convolution Process
- ❖ Type of Convolution

Filter Process

- Filtering is a way to extract certain part of a set of data, by eliminating the parts data is not desirable.
- **Aim:** making the image become more visible, or seem more obvious to be analyze.
 - Hold high frequency on imagea, smoothening images(smoothing)
 - Hold low frequency on images, clarifiesor detects edges in the image.

Filter Process

- **Frequency Domain.** Filtering the image in the frequency scopes will change the image into frequency domain, multiplying by a function of the frequency filter, transform back the results to spatially space scope.
- **Spatial Domain,** Spatial domain filtering is the process of manipulating a collection of pixels from an image to produce a new image.

Spatial Domain Filter

- Filtering the image of the space scope of spatial are doing convolution of image input $f(i,j)$ with the function of the filter $h(i,j)$, where the function of the filter that used to be simulated in the form of kernel discrete certain.

$$\mathbf{g(i,j) = h(i,j) \odot f(i,j)}$$

Convolution Process

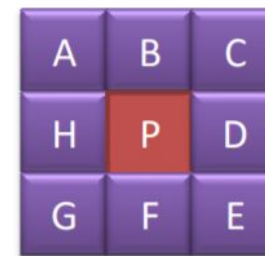
- Convolution is the sum of the multiplication of each point in the kernel by each point in the input function.
- Kernel operate as a shift in the function of the input image input $f(x)$. Total multiplication of each point on the function that is the result of convolution that is expressed by $g(x)$.
- The kernel uses the concept of neighboring pixels, where the kernel matrix is made with the assumption that the value of a pixel can be influenced by their neighbor pixels.



4-tetangga +



4-tetangga x

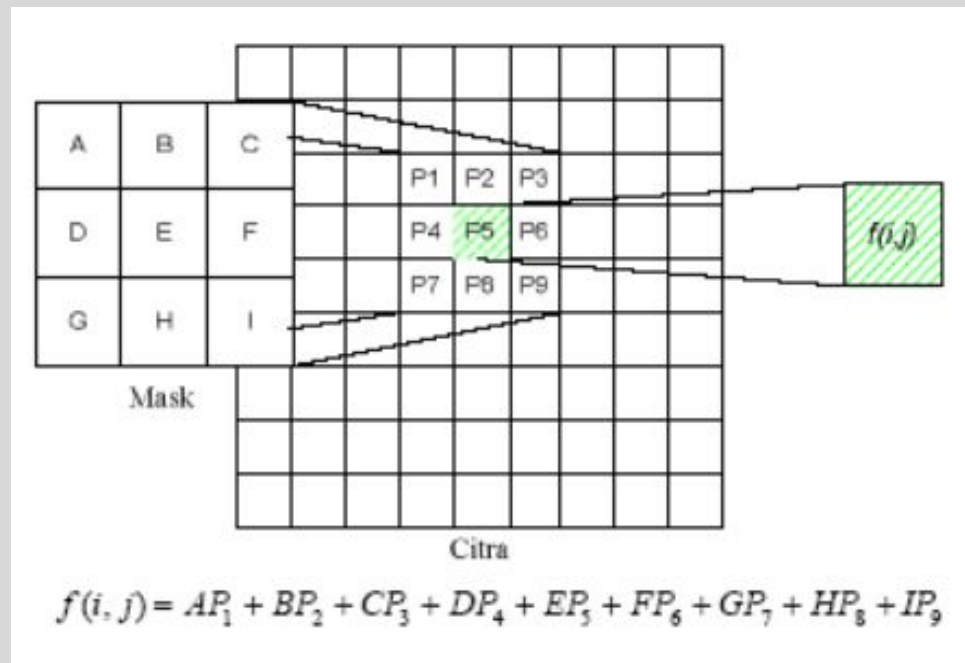


8-tetangga

Convolution Process

o

$$g(x, y) = f(x, y) \odot h(x, y) = \sum_{a=-\infty}^{\infty} \sum_{b=-\infty}^{\infty} f(a, b) g(x - a, y - b)$$



Convolution Process

Example

5	5	6	6
5	4	4	7
0	0	2	2
0	1	1	3

Image Input

$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

Kernel

$$\begin{aligned}g(2,2) &= 5 \times \frac{1}{4} + 5 \times \frac{1}{4} + 6 \times \frac{1}{4} + 5 \times \frac{1}{4} + 4 \times \frac{1}{4} + 4 \times \frac{1}{4} + 0 \times \frac{1}{4} + 0 \times \frac{1}{4} + 2 \times \frac{1}{4} \\ &= 1,25 + 1,25 + 1,5 + 1,25 + 1 + 1 + 0 + 0 + 0,5 = 7,75\end{aligned}$$

Convolution Process

Contoh

$$f(x, y) = \begin{bmatrix} 4 & 4 & 3 & 5 & 4 \\ 6 & 6 & 5 & 5 & 2 \\ 5 & 6 & 6 & 6 & 2 \\ 6 & 7 & 5 & 5 & 3 \\ 3 & 5 & 2 & 4 & 4 \end{bmatrix} \quad g(x, y) = \begin{bmatrix} 0 & -1 & 0 \\ -1 & *4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

catatan: tanda * menunjukkan posisi (0,0) dari kernel

Convolution Process

4	4	3	5	4
6	6	5	5	2
5	6	6	6	2
6	7	5	5	3
3	5	2	4	4



	3			

Hasil konvolusi = 3. Nilai ini dihitung dengan cara berikut:

$$(0 \times 4) + (-1 \times 4) + (0 \times 3) + (-1 \times 6) + (4 \times 6) + (-1 \times 5) + (0 \times 5) + (-1 \times 6) + (0 \times 6) = 3$$

4	4	3	5	4
6	6	5	5	2
5	6	6	6	2
6	7	5	5	3
3	5	2	4	4



	3	0		

Hasil konvolusi = 0. Nilai ini dihitung dengan dengan cara berikut:

$$(0 \times 4) + (-1 \times 3) + (0 \times 5) + (-1 \times 6) + (4 \times 5) + (-1 \times 5) + (0 \times 6) + (-1 \times 6) + (0 \times 6) = 0$$

4	4	3	5	4
6	6	5	5	2
5	6	6	6	2
6	7	5	5	3
3	5	2	4	4

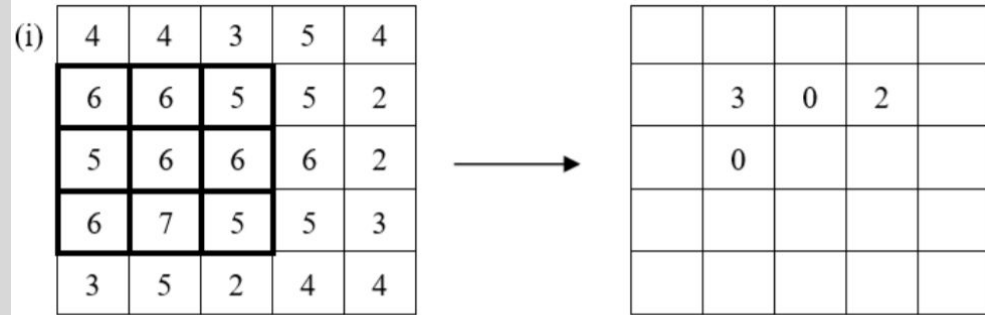


	3	0	2	

Hasil konvolusi = 2. Nilai ini dihitung dengan cara berikut:

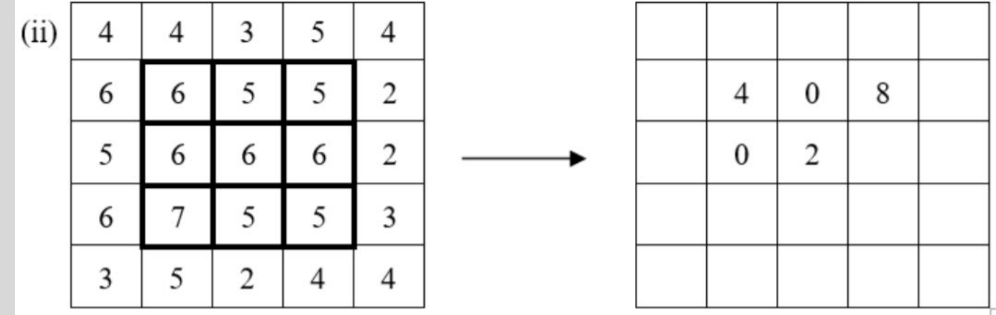
$$(0 \times 3) + (-1 \times 5) + (0 \times 4) + (-1 \times 5) + (4 \times 5) + (-1 \times 2) + (0 \times 6) + (-1 \times 6) + (0 \times 2) = 2$$

Convolution Process



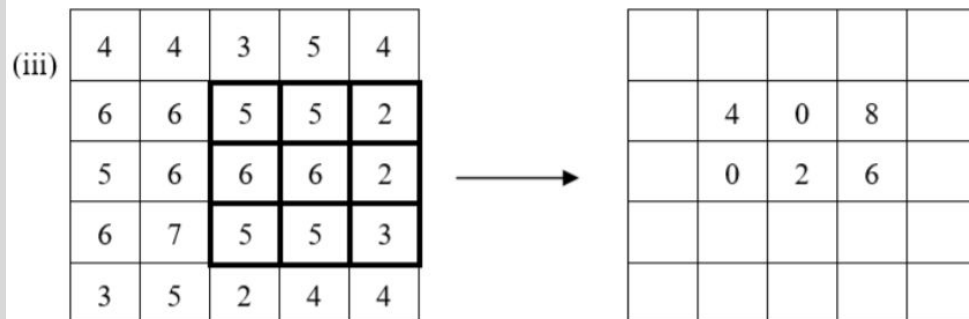
Hasil konvolusi = 0. Nilai ini dihitung dengan cara berikut:

$$(0 \times 6) + (-1 \times 6) + (0 \times 5) + (-1 \times 5) + (4 \times 6) + (-1 \times 6) + (0 \times 6) + (-1 \times 7) + (0 \times 5) = 0$$



Hasil konvolusi = 2. Nilai ini dihitung dengan cara berikut:

$$(0 \times 6) + (-1 \times 5) + (0 \times 5) + (-1 \times 6) + (4 \times 6) + (-1 \times 6) + (0 \times 7) + (-1 \times 5) + (0 \times 5) = 2$$



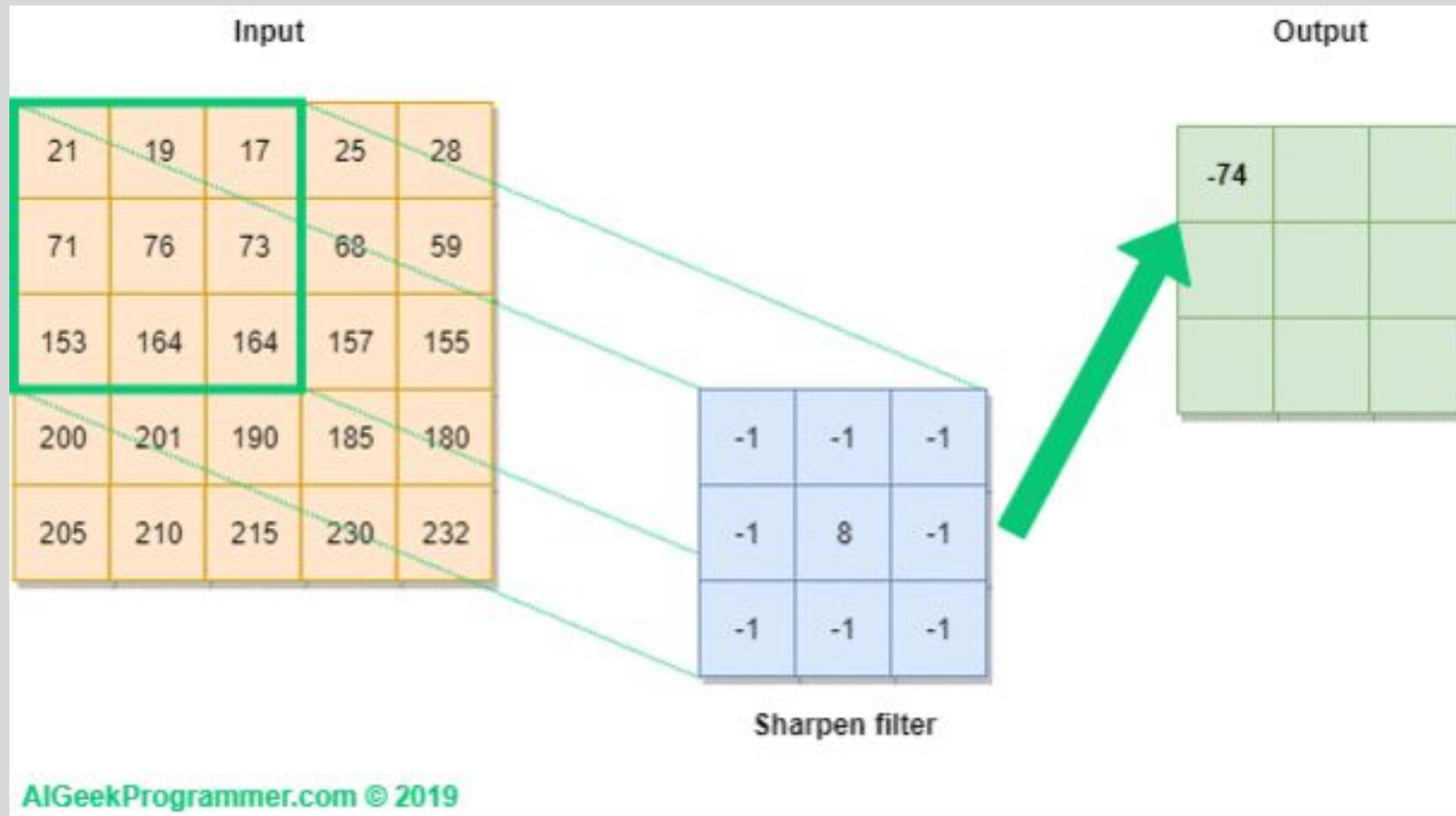
Hasil konvolusi = 6. Nilai ini dihitung dengan cara berikut:

$$(0 \times 5) + (-1 \times 5) + (0 \times 2) + (-1 \times 6) + (4 \times 6) + (-1 \times 2) + (0 \times 5) + (-1 \times 5) + (0 \times 3) = 6$$

Hasil

	4	0	8	
	0	2	6	
	6	0	2	

Convolution Process



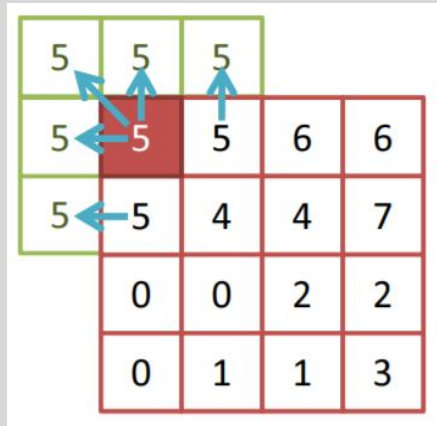
Convolution Process

Problems

4	4	3	5	4	?
6	6	5	5	2	?
5	6	6	6	2	?
6	7	5	5	3	
3	5	2	4	4	

Convolution Process

Solution

- Edge pixels are ignored, do not convolve.
 - Duplicate element of the image.
- 
- Elements that are marked with a "?" In Figure 3 , it is assumed that the value is 0 or some other constant , so that the edge pixel convolution can be carried out.

The diagram shows a 5x5 grid of cells. A 3x3 subgrid in the center is highlighted in red. A 3x3 subgrid in the top-left corner is highlighted in green. Blue arrows point from the green subgrid to the red subgrid, indicating a mapping from the green subgrid to the red subgrid.

5	5	5		
5	5	5	6	6
5	5	5	4	4
	0	0	2	2
	0	1	1	3

4	4	3	5	4
6	4	0	8	2
5	0	2	6	2
6	6	0	2	3
3	5	2	4	4

Diagram illustrating the first step of a 2D convolution operation:

- Image Matrix (6x6):**

0	0	0	0	0	0
0	105	102	100	97	96
0	103	99	103	101	102
0	101	98	104	102	100
0	99	101	106	104	99
0	104	104	104	100	98
- Kernel Matrix (3x3):**

0	-1	0
-1	5	-1
0	-1	0
- Output Matrix (6x6):**

320					
210	89	111			

The calculation for the top-left output value (320) is shown below:

$$\begin{aligned}
 &0 * 0 + 0 * 0 - 1 * 0 + 0 * 0 \\
 &+ 0 * -1 + 105 * 5 + 102 * -1 \\
 &+ 0 * 0 + 103 * -1 + 99 * 0 = 320
 \end{aligned}$$

Convolution Algorithm

```
For x = 0 to image_width-1
  For y = 0 image_height-1
    z(x,y) = 0
    For k1 = 0 to filter_width-1
      For k2 = 0 to filter_height-1
        z(x,y) = z(x,y) + H(k1,k2)*Img_pixel(x+k1, y+k2)
      next k2
    next k1
  next y
next x
```

Filter Type

- ❖ Sharpening (penajaman)
- ❖ Blurring (pengaburan)
- ❖ Emboss
- ❖ Edge detection (deteksi tepi)

Sharpening

Kernel for sharpening / sharpening the image using the principle that the intensity of the pixels centers should be strengthened in the direction that is opposite to its neighbors . Example of a kernel for image enhancement :

0	-1	0
-1	5	-1
0	-1	0

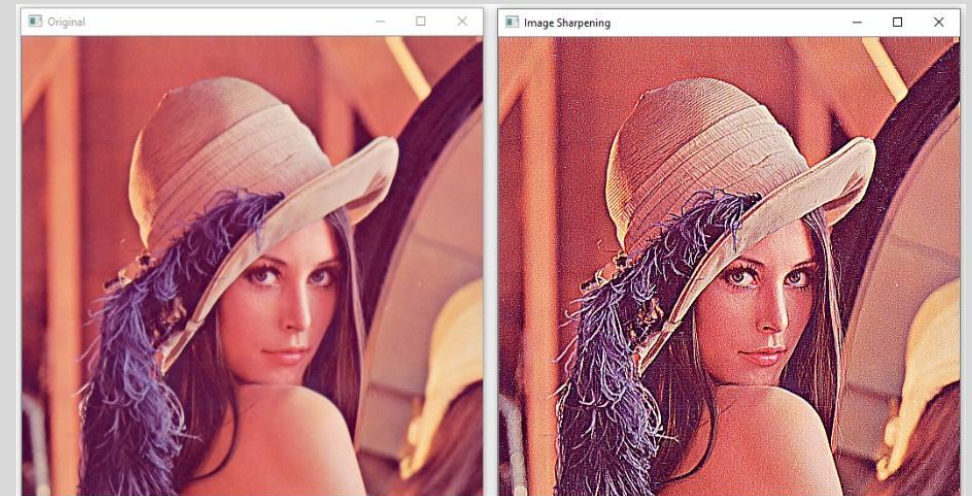
courtesy

<http://docs.gimp.org/en/plugin-in-convmatrix.html>

1	1	1
1	-8	1
1	1	1

courtesy

<http://www.imagesincontext.com/IICFeatures/convolution-filter.htm>

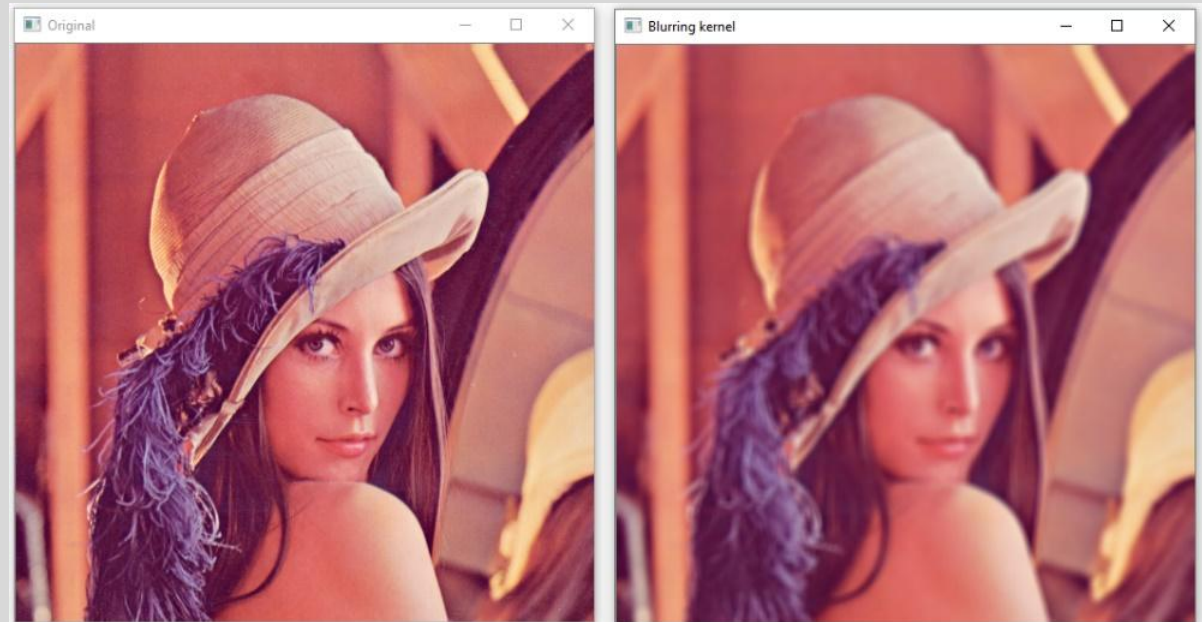


Blurring

The kernel for blurring / blurring uses the principle that the central pixel value should be made closer to neighboring pixels (reducing discrepancies). An example of a kernel for blurring images, one can use Gaussian Blurring:

$$\frac{1}{256} \begin{bmatrix} 1 & 4 & 6 & 4 & 1 \\ 4 & 16 & 24 & 16 & 4 \\ 6 & 24 & 36 & 24 & 6 \\ 4 & 16 & 24 & 16 & 4 \\ 1 & 4 & 6 & 4 & 1 \end{bmatrix}$$

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$



Emboss

The kernel for embossing is based on the principle of reinforcing edges in one particular direction , without losing any other color.

-2	-1	0
-1	1	1
0	1	2

courtesy

<http://docs.gimp.org/en/plugin-convmatrix.html>

1	0	0
0	0	0
0	0	-1

courtesy

<http://www.imagesincontext.com/IICFeatures/convolution-filter.htm>





THANK YOU!

